

| Department |                  |
|------------|------------------|
| PK         | <u>dept_id</u>   |
|            | name<br>location |

| Track |                 |
|-------|-----------------|
| PK    | <u>track_id</u> |
| FK    | dept_id<br>name |

| Track_Course |                 |
|--------------|-----------------|
| PK,FK1       | <u>track_id</u> |
| PK,FK2       | course_id       |

| Course |                                  |
|--------|----------------------------------|
| PK     | <u>course_id</u>                 |
|        | name<br>min_degree<br>max_degree |

| Student |                   |
|---------|-------------------|
| PK      | <u>student_id</u> |
|         | name<br>email     |

| Student_Track |                   |
|---------------|-------------------|
| PK,FK1        | <u>track_id</u>   |
| PK,FK2        | <u>student_id</u> |

| Instructor_Course |                      |
|-------------------|----------------------|
| PK,FK1            | <u>instructor_id</u> |
| PK,FK2            | <u>course_id</u>     |

| Instructor |                      |
|------------|----------------------|
| PK         | <u>instructor_id</u> |
|            | name<br>email        |
| FK         | dept_id              |

| Student_Answer |                        |
|----------------|------------------------|
| PK             | <u>id</u>              |
| FK1            | <u>student_exam_id</u> |
| FK2            | <u>question_id</u>     |
| FK3            | <u>option_id</u>       |

| Exam |                                                    |
|------|----------------------------------------------------|
| PK   | <u>exam_id</u>                                     |
| FK   | name<br>course_id<br>total_questions<br>created_at |

| Student_Exam |                                       |
|--------------|---------------------------------------|
| PK           | <u>id</u>                             |
| FK1          | <u>student_id</u>                     |
| FK2          | <u>exam_id</u>                        |
|              | start_time<br>end_time<br>total_grade |

| ModelAnswer |                            |
|-------------|----------------------------|
| PK,FK1      | <u>question_id</u>         |
| PK,FK2      | <u>option_id (correct)</u> |

| QuestionOption |                                            |
|----------------|--------------------------------------------|
| PK             | <u>option_id</u>                           |
| FK             | question_id<br>option_body<br>option_order |

| Exam_Question |                    |
|---------------|--------------------|
| PK,FK1        | <u>exam_id</u>     |
| PK,FK2        | <u>question_id</u> |
|               | order_number       |

| Question |                                              |
|----------|----------------------------------------------|
| PK       | <u>question_id</u>                           |
| FK       | course_id<br>question_body<br>type<br>points |